

Gabriel Guimarães Carvalho

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Education

Universidade Federal de Pernambuco – UFPE <i>PhD. Student in Computer Science</i>	Aug. 2019 – Aug. 2022 (interrupted) <i>Recife, BR</i>
Università di Roma “La Sapienza” <i>PhD. degree in Physics and Relativistic Astrophysics</i>	Feb. 2014 – Feb 2017 <i>Rome, IT</i>
Universidade Federal de Pernambuco – UFPE <i>M.Sc. degree in Mathematics</i>	2011 – 2013 <i>Recife, BR</i>
Universidade Federal de Pernambuco – UFPE <i>B.Sc. degree in Mathematics</i>	2008 – 2010 <i>Recife, BR</i>

Experience

Researcher Scientist - AI & Algorithms <i>Imec</i> Computer Vision and Deep Learning researcher	Jun. 2024 – <i>Antwerp, BE</i>
Senior Researcher Scientist <i>IDLab - UAntwerpen - Imec</i> Computer Vision and Deep Learning researcher	Jan. 2024 – May 2024 <i>Antwerp, BE</i>
Computer Vision Engineer <i>Moray</i> - Computer Vision and Image Processing - Data preprocessing and Data Analysis - Geographic Information System - Multispectral Drone and Satellite Image Processing - Pests and diseases detection in agricultural applications	Jul. 2022 – Nov. 2023 <i>São Paulo/Remote, BR</i>
Computer Vision Engineer <i>SpaceTime Labs</i> - Computer Vision and Image Processing - Data preprocessing and Data Analysis - Geographic Information System - Multispectral Drone and Satellite Image Processing	Jul. 2021 – Jul. 2022 <i>São Paulo/Remote, BR</i>
Ph.D. Researcher <i>Projeto CIn - Motorola</i> - Computational Photography - R&D of image processing and quality assessment tools - Deep Learning research and development - Data preprocessing and Data Analysis	Sep. 2019 – Jun. 2021 <i>Recife, BR</i>
Postdoctoral Researcher <i>Universidade Federal Rural de Pernambuco – UFRPE</i> Alternative gravitational theories and geometric methods in Physics	2017 – 2018 <i>Recife, BR</i>
Lecturer <i>Universidade Federal de Pernambuco – UFPE</i> - Calculus I, II, III - Analytic Geometry - Linear Algebra - Calculus in one complex variable	Nov 2017 – Jul 2019 <i>Recife, BR</i>
Lecturer <i>Universidade Federal de Pernambuco – UFPE</i> - Calculus I, II - Analytic Geometry	Apr 2012 – Jan 2014 <i>Recife, BR</i>

Technical Skills, Language Skills, and Interests

OS: Linux, macOS, Windows

Programming Languages: Python (proficient), C++ (familiar), Java (familiar), Matlab (familiar), Go (beginner)

Libraries: OpenCV, Pandas, Matplotlib, Numpy, Scipy, Scikit-Image, Scikit-Learn, PyTorch, PyTorch Lightning, Geopandas, Rasterio

Version Control: Git, DVC (familiar)

Tools: QGIS, Docker, Lightning Studio, MLFlow (familiar)

Cloud: AWS (familiar)

Writing: $\LaTeX$, Office

Languages: Portuguese (native), English (fluent, C1/C2), Italian (conversational, B2/C1), Norwegian (beginner, A1/A2)

Interests: Data Engineering, Machine Learning, Computer Vision and Image Processing, Mathematics, Physics, Travel, Music, Photography, Nature, Philosophy, Brazilian Jiu-Jitsu

Extracurricular

Projects: Personal Website and blog posts

Specialization: Deep Learning by DeepLearning.AI on Coursera. Certificate earned on July 12, 2020

Specialization: Mathematics for Machine Learning by Imperial College London on Coursera. Certificate earned on Dec. 12, 2020

Course: Machine Learning by Stanford University on Coursera. Certificate earned on Mar. 18, 2021

Course: Introduction to Quantum Information. Certificate earned on Feb. 3, 2025

Course: Advanced Computer Vision with TensorFlow. Certificate earned on Feb. 17, 2025

Specialization: IBM DevOps and Software Engineering on Coursera. Certificate earned on Apr. 28, 2025